

1	(a) (i) acceleration = change in velocity / time (taken) or acceleration = rate of change of velocity	B1	[1]
	(ii) a body continues at constant velocity unless acted on by a resultant force	B1	[1]
	(b) (i) distance is represented by the area under graph distance = $\frac{1}{2} \times 29.5 \times 3 = 44.3 \text{ m}$ (accept 43.5 m for 29 to 45 m for 30)	C1	
		A1	[2]
	(ii) resultant force = weight – frictional force frictional force increases with speed at start frictional force = 0 / at end weight = frictional force	B1	
		B1	
		B1	[3]
	(iii) 1. frictional force increases	B1	[1]
	2. frictional force (constant) and then decreases	B1	[1]
	(iv) 1. acceleration = $(v_2 - v_1) / t = (20 - 50) / (17 - 15)$ = $(-) 15 \text{ m s}^{-2}$	C1	
		A1	[2]
	2. $W - F = ma$ $W = 95 \times 9.81 (= 932)$ $F = (95 \times 15) + 932 = 2400 (2360) (2357) \text{ N}$	C1	
		C1	
		A1	[3]

2	(a) resistance = potential difference / current	B1	[1]
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